MF Analytics

A Quantitative Framework for Indian Equity Mutual Fund Analysis

Quantitative White Paper

March 26, 2026

Abstract

This white paper presents the complete quantitative framework underlying the MF Analytics platform for Indian equity mutual funds. We document the mathematical definitions, computational procedures, and interpretive guidelines for every metric exposed in the platform — spanning return analysis, risk measurement, risk-adjusted performance, benchmark-relative attribution, Systematic Investment Plan (SIP) modelling, and peer comparison via quartile ranking. All formulas are implemented verbatim in the platform's `analytics.py` engine, making this document both a technical reference and an investor education resource. Where appropriate we discuss the economic intuition behind each metric and note common misinterpretations.

Data sourced from AMFI India (NAVAll.txt) and Yahoo Finance (index prices). Risk-free rate: India 91-day T-bill, 6.5% p.a. Trading days per year: 252.

Contents

Introduction

Analysing Indian equity mutual funds rigorously requires navigating a landscape that is simultaneously rich in data and sparse in standardised tooling. The Association of Mutual Funds in India (AMFI) publishes daily Net Asset Values for over 2,000 equity schemes, yet retail investors and advisors alike typically rely on trailing return tables alone — an informational diet that obscures risk, benchmark sensitivity, and the experience of the median systematic investor.

The MF Analytics platform addresses this gap by computing a comprehensive set of quantitative metrics drawn from modern portfolio theory, empirical finance, and practitioner risk management. This white paper formalises the entire metric library with mathematical precision, documents design choices made during implementation, and provides interpretive context so that outputs can be translated into actionable investment insights.

Notation Conventions

Throughout this document we use the following notation.

Symbol	Meaning
P_t	NAV (or index level) on trading day t
$r_t = P_t/P_{t-1} - 1$	Simple daily return on day t
$\{r_t\}_{t=1}^{T}$	Time series of T daily returns
r_f	Annual risk-free rate (6.5% p.a.)
r_f^d	Daily equivalent: $(1 + r_f)^{1/252} - 1$
$N = 252$	Trading days per year
μ	Mean daily return
σ	Standard deviation of daily returns
β	Systematic risk relative to benchmark
R_p, R_b	Annualised fund and benchmark returns

Data Infrastructure

NAV Data Collection

The platform fetches the AMFI `NAVAll.txt` file to obtain the scheme catalogue (scheme code, name, SEBI category, ISIN). Historical daily NAVs are then retrieved from `api.mfapi.in` for each scheme and stored in a local SQLite database. Only *Direct Growth* plans are retained to ensure comparability: regular plans carry distributor commissions that inflate expense ratios and distort performance comparisons.

Benchmark Index Data

Index price series are fetched from Yahoo Finance using the `yfinance` library. The platform supports 16 confirmed-working tickers:

Index	Ticker	Segment
Nifty 50	`^NSEI`	Broad market large-cap
BSE Sensex	`^BSESN`	Broad market large-cap
Nifty Next 50	`^NSEMDCP50`	Large/mid transition
Nifty Midcap 50	`^NSMIDCP`	Mid-cap
Nifty Midcap 150	`NIFTYMIDCAP150.NS`	Mid-cap
Nifty Bank	`^NSEBANK`	Banking
Nifty Financial Svc	`NIFTY_FIN_SERVICE.NS`	Financial services
Nifty IT	`^CNXIT`	Technology
Nifty FMCG	`^CNXFMCG`	Consumer staples
Nifty Pharma	`^CNXPHARMA`	Healthcare
Nifty Infrastructure	`^CNXINFRA`	Infrastructure
Nifty Energy	`^CNXENERGY`	Energy
Nifty PSE	`^CNXPSE`	Public sector enterprises
Nifty Auto	`^CNXAUTO`	Automobiles & logistics
Nifty Realty	`^CNXREALTY`	Real estate
Nifty MNC	`^CNXMNC`	Multinational companies

Risk-Free Rate

All risk-adjusted metrics use a single annualised risk-free rate $r_f = 6.5\%$, approximating the average India 91-day Treasury Bill yield. The daily equivalent is computed as

$$r_f^d = (1 + r_f)^{1/252} - 1 \approx 0.025\%.$$

Fund Classification and Tagging

SEBI Categories

SEBI mandates that every open-ended equity mutual fund fall into exactly one regulatory category. The platform maps the following categories directly to a *style tag*:

SEBI Category	Style Tag
Equity Scheme – Large Cap Fund	Large Cap
Equity Scheme – Mid Cap Fund	Mid Cap
Equity Scheme – Small Cap Fund	Small Cap
Equity Scheme – Large & Mid Cap Fund	Large & Mid Cap
Equity Scheme – Multi Cap Fund	Multi Cap
Equity Scheme – Flexi Cap Fund	Flexi Cap
Equity Scheme – Focused Fund	Focused
Equity Scheme – Value Fund	Value
Equity Scheme – Contra Fund	Contra
Equity Scheme – ELSS	ELSS
Equity Scheme – Sectoral/Thematic	(resolved by name, see below)

Thematic Tagging via Rule Engine

Sectoral and thematic funds do not carry a standardised sub-category label. The platform resolves their tag by matching the lowercase fund name against an ordered list of keyword rules — the first match wins. Critical ordering constraints are:

1. **Business Cycle** and **Momentum** rules precede the **Quant & Factor** rule, because the AMC "Quant" names all its funds with the prefix "quant"; without priority ordering, the broad Quant keyword would absorb *quant Business Cycle Fund* and *quant Momentum Fund*.
2. Banking & Financial Services uses "banking", "financial services", "bfsi" but deliberately excludes "finserv" to avoid false matches on the BAJAJ FINSERV AMC name.

The 26 thematic tags, together with their recommended comparison benchmark, are listed in the configuration module `config.TAG_BENCHMARK`.

Return Metrics

Compound Annual Growth Rate (CAGR)

Definition 4.1 (CAGR)**.** *Let P_0 and P_T denote the NAV on the first and last day of the analysis period, and let D be the calendar-day span between them.*

$$\text{CAGR} = \left(\frac{P_T}{P_0}\right)^{365.25/D} - 1$$

The exponent uses 365.25 to account for leap years. CAGR is undefined when $D = 0$ or when $P_T/P_0 \leq 0$ (which cannot occur for NAVs).

Interpretation 4.1 (CAGR)**.** *CAGR gives the hypothetical constant annual growth rate that would turn P_0 into P_T in the same time span. It is a* geometric *average and therefore correctly reflects compounding. A fund with CAGR = 15% doubles in value approximately every 4.8 years (by the rule of 72).*

Absolute Return

$$R_{\text{abs}} = \frac{P_T}{P_0} - 1$$

Absolute return is period-length agnostic and should *not* be compared across funds with different holding periods. It is shown alongside CAGR for completeness.

Trailing Returns

For each standard lookback horizon Δ (1M, 3M, 6M, 1Y, 3Y, 5Y, 10Y), the platform locates the NAV on the date $t^* = T_{\text{end}} - \Delta$ and computes:

$$R_\Delta = \begin{cases} \dfrac{P_{T_{\text{end}}}}{P_{t^*}} - 1 & \text{if } \Delta < 1 \text{ year (simple return)} \\ \left(\dfrac{P_{T_{\text{end}}}}{P_{t^*}}\right)^{365.25/\Delta_d} - 1 & \text{if } \Delta \geq 1 \text{ year (annualised)} \end{cases}$$

where Δ_d is the lookback in calendar days. If fewer than 2 NAV data points exist in the lookback window, the value is reported as missing.

Rolling Returns

For a rolling window of W trading days, the annualised rolling return at time t is:

$$\hat{R}_t^{(W)} = \left(\frac{P_t}{P_{t-W}}\right)^{N/W} - 1$$

where $N = 252$. Rolling returns are used to visualise how a fund's return profile evolves over time and to construct the SIP return distribution.

Calendar Year Returns

The fund's NAV is resampled to year-end values. For year y:

$$R_y = \frac{P_{\text{Dec-end}(y)}}{P_{\text{Dec-end}(y-1)}} - 1$$

The first (and potentially last) calendar year is a partial year; returns are reported as-is without annualisation.

Risk Metrics

Annualised Volatility

Definition 5.1 (Annualised Volatility)**.**

$$\sigma = \hat{\sigma}_d \cdot \sqrt{N}$$

where $\hat{\sigma}_d = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})^2}$ *is the sample standard deviation of daily returns and* $N = 252$*.*

Interpretation 5.1 (Volatility)**.** *Volatility measures the dispersion of returns around their mean. A fund with* $\sigma = 20\%$ *experiences daily return movements with a typical magnitude of* $20\%/\sqrt{252} \approx 1.26\%$*. Crucially, volatility is* symmetric *— it penalises upside and downside fluctuations equally. This is a limitation addressed by the Sortino ratio and the Ulcer Index.*

Downside Deviation

Definition 5.2 (Downside Deviation)**.** *Let the daily minimum acceptable return be* $MAR_d = (1 + r_f)^{1/252} - 1$*. Define* $\delta_t = \min(r_t - MAR_d,\ 0)$*. Then:*

$$\sigma_D = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\delta_t^2} \cdot \sqrt{N}$$

Unlike standard deviation, downside deviation only penalises returns *below* the minimum acceptable threshold. It is the denominator of the Sortino ratio.

Maximum Drawdown

Definition 5.3 (Maximum Drawdown)**.** *Let $M_t = \max_{s \le t} P_s$ be the running all-time-high (ATH) at time t. The drawdown series is:*

$$d_t = \frac{P_t - M_t}{M_t} \le 0$$

Maximum drawdown is:

$$\text{MDD} = \min_t d_t$$

It is reported as a negative number (e.g. -0.35 means a 35% peak-to-trough decline).

Interpretation 5.2 (Maximum Drawdown)**.** *MDD answers: "what is the worst loss an investor could have experienced if they bought at the worst possible time?" For an equity fund MDD below -30% is common during bear markets; funds with MDD better than -20% in such periods often exhibit stronger risk management characteristics.*

Drawdown Analysis

Beyond the single worst event, the platform decomposes the entire drawdown history into *distinct episodes*:

1. An episode begins the first trading day $P_t < M_t$.
2. The **trough date** is the day of the minimum P_t.
3. The episode ends when the fund recovers to $P_t \ge M_{\text{start}-1}$ (a new ATH). If no recovery has occurred by the data end-date, the episode is marked "ongoing".
4. Three durations are recorded:
   - **Total duration** (days from start to recovery)
   - **Decline phase** (days from start to trough)
   - **Recovery phase** (days from trough to recovery)

Episodes are sorted by depth (worst first).

Ulcer Index

Definition 5.4 (Ulcer Index)**.**

$$\text{UI} = \sqrt{\frac{1}{T} \sum_{t=1}^{T} d_t^2}$$

where d_t is the drawdown series defined above.

The Ulcer Index (Martin & McCann, 1989) is the root-mean-square of all drawdown values. Because d_t^2 is large both when drawdowns are deep *and* when they persist over many days, UI penalises prolonged underwater periods more severely than MDD does.

Interpretation 5.3 (Ulcer Index)**.** *A UI of 5% is a rough baseline for a well-managed large-cap fund over a full market cycle. A fund with UI of 15% is delivering considerably more "ulcer-inducing" pain to its investors than MDD alone might suggest. Compare: two funds might share an MDD of -30%, but if one recovered in 60 days while the other spent two years underwater, their UIs will differ substantially.*

Average Drawdown

$$\bar{d} = \frac{1}{|\{t : d_t < 0\}|} \sum_{t:\, d_t<0} d_t$$

This is the mean drawdown over all days when the fund was below its ATH, reported as a negative fraction. It complements MDD by capturing the typical depth of drawdowns rather than only the worst.

Percentage of Time Underwater

$$\mathrm{PTU} = \frac{|\{t : d_t < 0\}|}{T}$$

The fraction of all trading days on which the fund was below its previous all-time high. A PTU of 40% means the fund was in drawdown 40% of the time — common for volatile sector funds.

Value at Risk (VaR)

Definition 5.5 (Historical VaR)**.** *At confidence level α (typically 95% or 99%), the one-day Historical VaR is the α-quantile of the empirical daily return distribution:*

$$\mathrm{VaR}_\alpha = -F^{-1}(1-\alpha)$$

where F is the empirical CDF of $\{r_t\}$. Because VaR is a loss quantity it is reported as a negative number in the platform (consistent with the drawdown convention).

Definition 5.6 (Parametric VaR)**.** *Assuming daily returns are normally distributed with mean μ and standard deviation σ:*

$$\mathrm{VaR}_\alpha^{param} = \mu + z_{1-\alpha}\,\sigma$$

where $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$ is the standard normal quantile (e.g. $z_{0.05} = -1.645$ for $\alpha = 0.95$).

The platform defaults to the historical method, which makes no distributional assumption. At least 30 daily observations are required for a meaningful estimate.

Interpretation 5.4 (VaR)**.** $\mathrm{VaR}_{95} = -2\%$ *means: on 95 out of 100 trading days, the fund's daily loss will not exceed 2%. Equivalently, there is a 5% chance of losing more than 2% in a single day. VaR does* not *describe the magnitude of losses in the tail — that is the role of CVaR.*

Conditional VaR (Expected Shortfall)

Definition 5.7 (CVaR / Expected Shortfall)**.**

$$\mathrm{CVaR}_\alpha = \mathbb{E}[r_t \mid r_t \le \mathrm{VaR}_\alpha] = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} r_t$$

where $\mathcal{T} = \{t : r_t \le \mathrm{VaR}_\alpha\}$ is the set of tail-loss days.

CVaR is coherent (satisfies sub-additivity) whereas VaR is not. It answers: "given that we are in the worst $(1-\alpha)$% of outcomes, what is the *average* loss?"

Return Skewness and Kurtosis

Skewness quantifies the asymmetry of the return distribution:

$$\text{Skew} = \frac{\mathbb{E}[(r_t - \mu)^3]}{\sigma^3}$$

Negative skewness (left tail heavier) is common in equity funds — large losses are more frequent than large gains of the same magnitude.

Excess kurtosis quantifies tail heaviness relative to the normal distribution:

$$\text{Kurt} = \frac{\mathbb{E}[(r_t - \mu)^4]}{\sigma^4} - 3$$

Positive excess kurtosis (leptokurtosis) implies more frequent extreme outcomes (both gains and losses) than a normal distribution would predict. Most equity return series exhibit positive excess kurtosis of 2–6.

Risk-Adjusted Return Metrics

Sharpe Ratio

Definition 6.1 (Sharpe Ratio).

$$S = \frac{R_p - r_f}{\sigma}$$

where R_p is the fund's CAGR, $r_f = 6.5\%$, and σ is the annualised volatility.

Interpretation 6.1 (Sharpe Ratio). *The Sharpe ratio measures the excess return per unit of total risk. A value above 1.0 is considered good; above 2.0 is excellent for an equity fund over a full market cycle. Values below 0.5 suggest the fund is not adequately compensating investors for the volatility they bear.*

Key limitation: because σ is symmetric, a fund that achieves high returns with high upside volatility (many large positive days) can have an inflated Sharpe. This is addressed by the Sortino and Martin ratios.

Sortino Ratio

Definition 6.2 (Sortino Ratio).

$$\textit{Sortino} = \frac{R_p - r_f}{\sigma_D}$$

where σ_D is the annualised downside deviation.

Interpretation 6.2 (Sortino Ratio). *By replacing σ with σ_D, the Sortino ratio only penalises* negative *deviations from the minimum acceptable return. A fund with strongly right-skewed returns (many moderate positives, rare large negatives) will have Sortino $\gg$ Sharpe, correctly reflecting the investor's actual experience. For most equity funds the Sortino ratio is approximately 1.5–2× the Sharpe ratio.*

Calmar Ratio

Definition 6.3 (Calmar Ratio)**.**

$$Calmar = \frac{R_p}{|\text{MDD}|}$$

The Calmar ratio (Young, 1991) relates annualised return to the worst historical drawdown. It is popular among hedge fund practitioners because it uses a concrete worst-case loss metric rather than a statistical quantity. A Calmar ratio of 0.5 means a fund earned half its worst drawdown in annualised return — e.g. 10% CAGR against a 20% MDD.

Martin Ratio (Ulcer Performance Index)

Definition 6.4 (Martin Ratio)**.**

$$Martin = \frac{R_p - r_f}{\text{UI}}$$

where UI *is the Ulcer Index.*

Interpretation 6.3 (Martin Ratio)**.** *Proposed by Peter Martin in 1987 (and commercially popularised as the* Ulcer Performance Index*), the Martin ratio is a Sharpe-like ratio that substitutes the Ulcer Index for volatility. Because UI is sensitive to both depth and duration of drawdowns, the Martin ratio penalises funds that spend long periods recovering from losses — even if those losses were individually modest.*

Compared to Sharpe: a fund that declines slowly but persistently scores worse on Martin than Sharpe. A fund that recovers rapidly from a sharp crash scores better on Martin than Sharpe.

Percentage of Positive Rolling Periods

$$\text{P}^+_{(W)} = \frac{|\{t : \hat{R}_t^{(W)} > 0\}|}{|\{t : \hat{R}_t^{(W)} \text{ defined}\}|}$$

For a 1-year rolling window ($W = 252$), this answers: "what fraction of all possible 1-year holding periods delivered a positive return?" A value above 90% indicates the fund rarely disappoints investors who hold for a full year.

Benchmark-Relative Metrics

Let $\{r_t^f\}$ and $\{r_t^b\}$ denote the daily returns of the fund and benchmark, respectively, on their common trading days. All benchmark metrics require at least 30 aligned observations.

Beta

Definition 7.1 (Beta)**.**

$$\beta = \frac{\text{Cov}(r^f, r^b)}{\text{Var}(r^b)}$$

Beta measures the fund's systematic sensitivity to benchmark movements. A beta of 1.2 implies the fund tends to move 1.2% for every 1% benchmark move (leveraged market exposure). Beta below 1 implies defensive characteristics.

The sample estimator used is:

$$\hat{\beta} = \frac{\sum_{t=1}^{T}(r_t^f - \bar{r}^f)(r_t^b - \bar{r}^b)}{\sum_{t=1}^{T}(r_t^b - \bar{r}^b)^2}$$

Jensen's Alpha

Definition 7.2 (Jensen's Alpha)**.**

$$\alpha = R_p - [r_f + \beta\,(R_b - r_f)]$$

where R_p and R_b are the fund's and benchmark's annualised CAGR.

Interpretation 7.1 (Jensen's Alpha)**.** *Alpha is the return earned in excess of what the Capital Asset Pricing Model (CAPM) predicts given the fund's market exposure. Positive alpha indicates skill: the fund manager added value beyond simply taking benchmark risk.*

Numerically, $\alpha = 3\%$ means the fund delivered 3 percentage points per year more than a passive strategy with the same beta. The platform computes alpha using the full-period CAGR for both the fund and benchmark, rather than OLS regression intercepts, to keep the interpretation consistent with the other return metrics.

Tracking Error

Definition 7.3 (Tracking Error)**.**

$$\mathrm{TE} = \mathrm{Std}(r_t^f - r_t^b) \cdot \sqrt{N}$$

Tracking error is the annualised standard deviation of daily active returns (fund minus benchmark). For index funds, TE should be below 0.5%; for actively managed funds TE typically ranges from 3% to 10%.

Information Ratio

Definition 7.4 (Information Ratio)**.**

$$\mathrm{IR} = \frac{R_p - R_b}{\mathrm{TE}}$$

The Information Ratio (IR) measures the consistency of outperformance: it rewards a high active return ($R_p - R_b$) but penalises inconsistency (high TE). An IR above 0.5 over a 3-year period is widely considered evidence of skill. The IR is often called the "Sharpe ratio of active management".

Treynor Ratio

Definition 7.5 (Treynor Ratio)**.**

$$T = \frac{R_p - r_f}{\beta}$$

The Treynor ratio replaces total volatility with systematic risk (β). It is most useful when comparing funds that are part of a well-diversified portfolio, where idiosyncratic risk is already diversified away.

Upside and Downside Capture Ratios

Capture ratios are computed on *monthly* returns. Let $\mathcal{U} = \{m : r^b_m > 0\}$ be the set of months in which the benchmark was positive ("up months"), and $\mathcal{D} = \{m : r^b_m < 0\}$ the set of "down months".

Definition 7.6 (Upside Capture Ratio).

$$\text{UCR} = \frac{G^f_{\mathcal{U}}}{G^b_{\mathcal{U}}} \times 100$$

where $G^f_{\mathcal{U}} = \left(\prod_{m\in\mathcal{U}}(1 + r^f_m)\right)^{1/|\mathcal{U}|} - 1$ *is the geometric mean monthly return of the fund in up months, and similarly for* $G^b_{\mathcal{U}}$.

Definition 7.7 (Downside Capture Ratio).

$$\text{DCR} = \frac{G^f_{\mathcal{D}}}{G^b_{\mathcal{D}}} \times 100$$

Interpretation 7.2 (Capture Ratios). *An ideal fund has UCR* > 100 *(participates fully in rallies) and DCR* < 100 *(cushions declines). A fund with UCR* $= 110$ *and DCR* $= 80$ *captures 110% of up-market gains but only 80% of down-market losses — an asymmetry highly favourable to long-term compounding. Funds in the upper-left quadrant of a UCR vs DCR scatter plot are the most desirable.*

Note: capture ratios below 3 months of data are unreliable and marked as missing.

Batting Average

Definition 7.8 (Batting Average).

$$\text{BA} = \frac{|\{m : r^f_m > r^b_m\}|}{M}$$

where M *is the total number of months with data for both the fund and benchmark.*

Batting average measures the frequency of outperformance, without regard to magnitude. A BA of 0.55 means the fund beat its benchmark in 55% of months. A high BA combined with a high UCR and low DCR is a strong signal of consistent active management quality.

Rolling Metrics

All rolling metrics use a backward-looking window of W trading days (configurable: 126, 252, 504, 756, 1260 for 6M, 1Y, 2Y, 3Y, 5Y).

Rolling Volatility

$$\sigma^{(W)}_t = \hat{\sigma}(\{r_{t-W+1}, \ldots, r_t\}) \cdot \sqrt{N}$$

Rolling Sharpe Ratio

$$S^{(W)}_t = \frac{\hat{R}^{(W)}_t - r_f}{\sigma^{(W)}_t}$$

where $\hat{R}_t^{(W)} = (P_t/P_{t-W})^{N/W} - 1$ is the annualised return over the trailing window.

Rolling Sortino Ratio

$$\text{Sortino}_t^{(W)} = \frac{\hat{R}_t^{(W)} - r_f}{\hat{\sigma}_{D,t}^{(W)}}$$

where $\hat{\sigma}_{D,t}^{(W)}$ is the rolling downside deviation:

$$\hat{\sigma}_{D,t}^{(W)} = \sqrt{\frac{1}{W}\sum_{s=t-W+1}^{t} \min(r_s - r_f^d, 0)^2} \cdot \sqrt{N}$$

Rolling Alpha and Beta

$$\hat{\beta}_t^{(W)} = \frac{\sum_{s=t-W+1}^{t}(r_s^f - \bar{r}^f)(r_s^b - \bar{r}^b)}{\sum_{s=t-W+1}^{t}(r_s^b - \bar{r}^b)^2}$$

$$\hat{\alpha}_t^{(W)} = \hat{R}_t^{(W),f} - \left[r_f + \hat{\beta}_t^{(W)}\left(\hat{R}_t^{(W),b} - r_f\right)\right]$$

Rolling alpha and beta reveal structural changes in a manager's risk positioning over time. A declining beta over time may signal a defensive tilt, while rising alpha may indicate an improving stock-selection process.

SIP Analysis and XIRR

Systematic Investment Plans (SIPs) are the dominant mode of retail mutual fund investing in India. Evaluating a fund purely on point-to-point CAGR ignores the cost averaging and path-dependency inherent in monthly investments.

XIRR: Extended Internal Rate of Return

XIRR is the standard of AMFI, SEBI disclosures, and Excel for evaluating irregular cashflow series. It solves for the annualised discount rate r that sets the Net Present Value of all cashflows to zero.

Definition 9.1 (XIRR). *Given cashflows $C_0, C_1, \ldots, C_n$ at dates $t_0 < t_1 < \cdots < t_n$ (with t_0 as the reference date), XIRR solves:*

$$\sum_{i=0}^{n} \frac{C_i}{(1+r)^{(t_i - t_0)/365.25}} = 0$$

Sign convention (Excel XIRR): investments are negative *(cash out of the investor's pocket); redemptions are* positive *(cash received).*

The platform uses `scipy.optimize.brentq` with bracket $r \in (-0.9999,\ 200.0)$ to accommodate the very high XIRRs achievable by short-term SIPs in bull markets.

SIP Return Computation

For a Y-year SIP ending on the last available NAV date:

1. Generate $Y \times 12$ month-start dates ending at the last NAV date.
2. For each month-start date m_i, find the NAV on or immediately after m_i in the price series.
3. Invest Rs.1 at that NAV, purchasing $u_i = 1/\text{NAV}_i$ units:
$$C_i = -1, \quad \text{units accumulated } \to U = \sum_i u_i$$
4. At the end of the period, redeem all units at the final NAV:
$$C_{\text{redemption}} = +U \times P_T$$
5. Solve $\sum_i C_i/(1+r)^{(t_i - t_0)/365.25} = 0$ for r.

SIP XIRR is computed for horizons $Y \in \{1, 3, 5, 10\}$ years.

Interpretation 9.1 (SIP XIRR)**.** *SIP XIRR captures cost averaging: if a fund fell sharply midway through the SIP and then recovered, more units were accumulated at low prices, and the XIRR can exceed the equivalent lump-sum CAGR. Conversely, a fund that rallied early and then corrected will show SIP XIRR below its CAGR. For most long-term equity funds, 5-year and 10-year SIP XIRR is a robust estimate of the "real investor experience".*

Rolling SIP Distribution

To assess *consistency* of SIP outcomes, the platform evaluates XIRR for every possible Y-year SIP starting date in the fund's history. This produces a distribution of XIRRs indexed by start date. Summary statistics of interest:

- **Percentage positive**: fraction of all Y-year SIP windows with XIRR > 0. Values below 90% are concerning for a 3-year horizon and should be near 100% for a 10-year horizon in a mature equity fund.
- **Worst XIRR**: the SIP window that started at the most unfavourable time (typically right before a major crash).
- **Median XIRR**: more robust than mean to outlier windows.

Peer Comparison and Quartile Ranking

Metrics Comparison Table

The platform constructs an $M \times F$ matrix where M metrics are rows and F funds (within a tag or user-selected group) are columns. Each cell (m, f) contains $v_{m,f}$, the value of metric m for fund f. Values are formatted as percentages for return/risk metrics and as plain decimals for ratios.

Within-Category Quartile Ranking

For each metric row m, funds are ranked against their peers:

1. Let $\mathcal{F}_m = \{f : v_{m,f} \text{ is finite}\}$ be the set of funds with valid values.
2. Determine the ranking direction:

$$\text{ascending} = \begin{cases} \text{True} & \text{if } m \in \mathcal{L} \quad \text{(lower is better)} \\ \text{False} & \text{otherwise (higher is better)} \end{cases}$$

   where $\mathcal{L}$ = {volatility, ulcer_index, average_drawdown, pct_time_underwater, tracking_error, var_95, var_99, cvar_95, cvar_99, worst_drawdown_depth, worst_drawdown_duration, downside_capture}.
3. Assign rank $\rho_f = \text{rank}(v_{m,f})/|\mathcal{F}_m|$ (using minimum rank for ties).
4. Classify:

$$Q_f = \begin{cases} \text{Q1} & \rho_f \le 0.25 \\ \text{Q2} & 0.25 < \rho_f \le 0.50 \\ \text{Q3} & 0.50 < \rho_f \le 0.75 \\ \text{Q4} & \rho_f > 0.75 \end{cases}$$

Q1 (best quartile) is rendered in green; Q4 (worst quartile) in red.

Interpretation 10.1 (Quartile Rankings)**.** *Quartile rankings provide the answer to the question that raw numbers cannot: "is this fund good* relative to its peers*?" A large-cap fund with a Sharpe of 0.7 may look mediocre in absolute terms but could be Q1 within its category if most large-cap funds have Sharpe below 0.5. Conversely, a sector fund with Sharpe 1.5 might be Q3 if it operates in a high-return, low-volatility niche.*

Users should look for funds that are consistently Q1 or Q2 across multiple *metrics rather than exceptional in one metric and poor in others.*

Return Colour Coding

The heatmap of trailing returns uses a signed per-column gradient:

$$\text{scaled}_{f,\Delta} = \frac{R_{f,\Delta}/\max_f |R_{f,\Delta}| + 1}{2} \in [0,1]$$

This maps the range $[-\max|\cdot|, 0, +\max|\cdot|]$ to $[0, 0.5, 1]$ in the `RdYlGn` colormap: red for negative returns, yellow for zero, and green for positive returns — regardless of whether all values happen to be positive or all negative. The per-column normalisation ensures that both short (1M) and long (5Y) columns use the full colour range, making it easy to spot anomalies at any horizon.

Implementation Notes

Data Quality

- Dates in `mfapi.in` are in DD-MMM-YYYY format (e.g. "15-Jan-2020") and are converted to ISO 8601 before storage.
- Duplicate NAV entries for the same (scheme_code, date) are silently deduplicated (last-write wins).
- The benchmark series and fund NAV series are aligned on common dates before any benchmark-relative computation.
- All metrics require a minimum of 30 daily observations; fewer than 252 points disables rolling 1-year metrics.

Numerical Stability

- The XIRR solver uses Brent's method with bracket $r \in (-0.9999, 200.0)$ and tolerance 10^{-6}. A bracket of 200 (20,000% p.a.) is required because a 1-year SIP into a 4× equity rally can legitimately produce XIRRs above 100%.
- Rolling beta denominators of zero (flat benchmark windows) are replaced by NaN.
- All downstream callers test for NaN and $\pm\infty$ using `pd.notna(v) and np.isfinite(v)` rather than `np.isnan(v)`, which raises `TypeError` on `None`.

Tagger Rule Ordering

The thematic keyword rule engine uses a first-match-wins strategy. The ordering constraint that is non-obvious and critical: *Business Cycle* and *Momentum* rules must appear before the *Quant & Factor* rule in the ordered list. Quant Asset Management Company names all its funds "quant XYZ Fund", so any keyword containing "quant" would absorb "quant Business Cycle Fund" and "quant Momentum Fund" into the wrong bucket. The solution is to restrict Quant & Factor keywords to "quant fund", "quantamental", etc. (not bare "quant"), and ensure Business Cycle and Momentum patterns match first.

Common-Period Calculation

When displaying all funds in a tag on a single growth chart, a common analysis window is required. Naively using $[\max(\text{starts}), \min(\text{ends})]$ fails when a tag contains a newly-launched fund (start $\approx$ today) alongside discontinued funds (end in the past), yielding an empty or inverted window.

The platform uses:

$$t_{\text{end}} = \max(\text{all end dates}), \qquad t_{\text{start}} = P_{80}(\text{sorted start dates})$$

where P_{80} is the 80th percentile of start dates. This excludes the 20% most-recently-launched funds from constraining the window, while still representing the majority of funds over a meaningful history.

Summary Metric Reference

Table 1: Complete Metric Reference

Metric	Formula / Definition	Interpretation
CAGR	$(P_T/P_0)^{365.25/D} - 1$	Annualised total return; primary headline metric.
Trailing Returns	Period-end vs period-start; annualised for $\geq$1Y	Snapshot of recent performance at standard horizons.
Volatility	$\hat{\sigma}_d \cdot \sqrt{252}$	Total risk; symmetric — penalises up and down moves.
Downside Dev.	$\sqrt{\frac{1}{T}\sum \min(r_t - r_f^d, 0)^2} \cdot \sqrt{252}$	Risk below MAR; denominator of Sortino ratio.
Max Drawdown	$\min_t (P_t - M_t)/M_t$	Worst peak-to-trough decline; tail-risk proxy.
Ulcer Index	$\sqrt{\frac{1}{T}\sum d_t^2}$	RMS of all drawdowns; penalises depth *and* duration.
Average Drawdown	Mean of d_t where $d_t < 0$	Typical drawdown depth while underwater.
% Time Underwater	$\lvert\{t : d_t < 0\}\rvert/T$	Fraction of days below all-time high.
VaR (95%, 99%)	α-quantile of $\{r_t\}$ (historical)	Daily loss not exceeded on $(1-\alpha)$ of days.
CVaR (95%, 99%)	$\mathbb{E}[r_t \mid r_t \leq \mathrm{VaR}_\alpha]$	Expected loss in the worst $(1-\alpha)$% of days.
Skewness	$\mathbb{E}[(r_t - \mu)^3]/\sigma^3$	Negative = fatter left tail (more frequent large losses).
Kurtosis	$\mathbb{E}[(r_t - \mu)^4]/\sigma^4 - 3$	Positive = heavier tails than normal distribution.
Sharpe Ratio	$(R_p - r_f)/\sigma$	Excess return per unit of total risk. > 1 is good.
Sortino Ratio	$(R_p - r_f)/\sigma_D$	Excess return per unit of downside risk.
Calmar Ratio	$R_p/\lvert\mathrm{MDD}\rvert$	Return vs worst loss; higher is better.
Martin Ratio	$(R_p - r_f)/\mathrm{UI}$	Sharpe variant using Ulcer Index; penalises prolonged pain.
% Positive 1Y	Fraction of 1Y rolling windows with > 0 return	Consistency of annual positive returns.
Beta	$\mathrm{Cov}(r^f, r^b)/\mathrm{Var}(r^b)$	Systematic exposure to benchmark; 1 = market-neutral risk.
Jensen's Alpha	$R_p - [r_f + \beta(R_b - r_f)]$	Excess return beyond CAPM; positive = manager skill.
Tracking Error	$\mathrm{Std}(r^f - r^b) \cdot \sqrt{252}$	Variability of active returns; high for concentrated bets.

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Metric	Formula / Definition	Interpretation
Information Ratio	$(R_p - R_b)/\mathrm{TE}$	Consistency of outperformance; > 0.5 signals skill.
Treynor Ratio	$(R_p - r_f)/\beta$	Return per unit of *systematic* risk only.
Upside Capture	$(G^f_{\mathcal{U}}/G^b_{\mathcal{U}}) \times 100$	> 100: fund rises more than benchmark in up months.
Downside Capture	$(G^f_{\mathcal{D}}/G^b_{\mathcal{D}}) \times 100$	< 100: fund falls less than benchmark in down months.
Batting Average	Fraction of months fund > benchmark	Frequency of monthly outperformance.
SIP XIRR 1Y/3Y/5Y/10Y	Brent's method on monthly Rs.1 SIP cashflows	Annualised return of the systematic investor.

Common Misinterpretations and Caveats

1. **Short-period CAGR is misleading.** A fund with 6 months of history showing CAGR 40% simply experienced a bull market at launch. Require at least 3 years of history before drawing conclusions about manager skill.
2. **Alpha without statistical testing.** The platform reports Jensen's alpha as a point estimate. A positive alpha does not necessarily reflect skill — it could be luck, style exposure, or small-sample noise. For robustness, look for consistent positive alpha across multiple sub-periods.
3. **Beta is not constant.** Fund managers actively change their market exposure. A fund's full-history beta may differ substantially from its recent 1-year rolling beta. Always consult the rolling beta chart alongside the static value.
4. **VaR does not bound the loss.** $\mathrm{VaR}_{95} = -2\%$ means losses exceed 2% in 5% of days — but on those bad days, losses could be -5%, -10%, or worse. CVaR (Expected Shortfall) provides the expected loss in that tail.
5. **Tracking error and alpha are not independent.** A manager with $\mathrm{TE} = 0.5\%$ cannot have large alpha — they are effectively running an index fund. High TE is the *necessary* (but not sufficient) condition for large positive alpha.
6. **SIP XIRR depends on the end date.** If the fund's current NAV is at a cyclical peak, the SIP XIRR will be inflated; if it is in a trough, it will be depressed. The rolling SIP distribution is a more robust characterisation of the likely investor experience.
7. **Quartile ranks are peer-relative.** A Q1 fund in a poor category may still be an absolute underperformer. Always examine absolute values alongside peer ranks.
8. **Direct vs Regular plan comparison.** The platform exclusively uses Direct Growth plans. Comparing results against industry data that mixes Direct and

Regular plans will show an apparent outperformance equal approximately to the TER differential (typically 0.5–1.5% p.a.).

Platform Architecture and Implementation

This section documents the full technical architecture of the MF Analytics platform — covering module design, database schema, data pipeline, application layer, tagging engine, caching strategy, and deployment. Together with the metric definitions in earlier sections, it constitutes a complete specification of the system.

High-Level Architecture

The platform is a single-host, Python-based application with four logical layers:

Layer	Components
Presentation	Streamlit web application (`app.py`) — multi-page UI rendered in the browser
Analytics	Pure-Python computation engine (`analytics.py`) — stateless functions operating on `pd.Series` inputs
Data Access	SQLite database (`mf_data.db`) accessed through `db.py`; fetchers in `fetcher.py` bridge external APIs to the local store
Configuration	`config.py` — all environment-specific constants (paths, URLs, rates, benchmarks) in one place

Design principle. The analytics layer is fully decoupled from both the database and the UI: `analytics.py` accepts plain `pd.Series` arguments and returns plain Python dicts or Series. This makes every metric independently testable and reusable outside Streamlit.

Module dependency graph

```
config.py
  ↑ imported by all modules
db.py   (depends on:  config)
fetcher.py   (depends on:  config, db)
tagger.py   (depends on:  db)
analytics.py   (depends on:  config only)
update.py   (depends on:  db, fetcher)
app.py   (depends on:  config, db, fetcher, tagger, analytics)
```

Source Code Modules

Table 2: Module inventory

Module	LOC	Responsibility
`config.py`	132	All configuration constants: file paths, API endpoints, rate limits, risk-free rate, trading-days constant, benchmark ticker map, NSE direct-index map, tag-to-benchmark map, rolling windows, VaR confidence levels.
`db.py`	260	SQLite schema definition, WAL-mode connection context manager, upsert/bulk-insert helpers, query helpers for NAV and benchmark retrieval, update-log management, and DB statistics.
`fetcher.py`	511	Data acquisition from three external sources: AMFI `NAVAll.txt` (scheme list), `mfapi.in` REST API (full NAV history per scheme), `yfinance` (benchmark index prices), and the NSE daily archive CSV files (smallcap/microcap indices not available on yfinance).
`tagger.py`	317	Two-pass rule engine that assigns sector/style tags to every fund. Manages the `fund_tags` table, exposes query helpers for tag-based filtering, and supports per-fund manual tag override.
`analytics.py`	765	Complete metric library: trailing returns, CAGR, volatility, drawdown series, Sharpe/Sortino/Calmar/Martin/Ulcer ratios, alpha, beta, R^2, tracking error, information ratio, capture ratios, rolling variants of each, SIP XIRR, calendar year returns, peer comparison table.
`update.py`	113	Standalone daily updater script. Orchestrates scheme-list refresh, incremental NAV download, and benchmark refresh. Supports one-shot and scheduled (`cron`-friendly) execution modes.
`app.py`	2056	Streamlit application. Five pages (Dashboard, Fund Analysis, Fund Comparison, Sector & Style, Data Management), module-level `@st.cache_data` wrappers, sidebar navigation, and all interactive visualisations.

Database Schema

All persistent state lives in a single SQLite file (`mf_data.db`) co-located with the source code. WAL journal mode and `PRAGMA synchronous=NORMAL` are set on every connection for safe concurrent reads while a write is in progress.

Table: `funds`

```
CREATE TABLE funds (
    scheme_code   INTEGER PRIMARY KEY,   -- AMFI unique scheme code
    scheme_name   TEXT    NOT NULL,
```

```
    scheme_category TEXT,                    -- e.g. "Equity Scheme - Large
       Cap Fund"
    scheme_type    TEXT,                     -- "Open Ended Schemes" etc.
    fund_house     TEXT,                     -- derived from scheme_name (
       first token)
    isin_growth    TEXT,                     -- ISIN of growth variant
    isin_div       TEXT                      -- ISIN of IDCW/dividend variant
);
CREATE INDEX idx_funds_category ON funds(scheme_category);
```

Listing 1: funds — scheme master

The table is populated from AMFI `NAVAll.txt` on every scheme-list refresh. Upserts use `ON CONFLICT ... DO UPDATE SET` with `COALESCE` so that non-null fields are never overwritten by a null.

Table: `nav_history`

```
CREATE TABLE nav_history (
    scheme_code  INTEGER NOT NULL,
    date         TEXT    NOT NULL,    -- ISO-8601: "YYYY-MM-DD"
    nav          REAL    NOT NULL,
    PRIMARY KEY (scheme_code, date),
    FOREIGN KEY (scheme_code) REFERENCES funds(scheme_code)
);
CREATE INDEX idx_nav_date   ON nav_history(date);
CREATE INDEX idx_nav_scheme ON nav_history(scheme_code);
```

Listing 2: nav_history — daily NAV time series

Records are inserted with `INSERT OR IGNORE` so incremental downloads never duplicate data. Date strings are stored in ISO-8601 lexicographic order so `MAX(date)` and range queries work without date parsing.

Table: `benchmark_data`

```
CREATE TABLE benchmark_data (
    index_name   TEXT NOT NULL,    -- display name, e.g. "Nifty 50"
    date         TEXT NOT NULL,    -- ISO-8601: "YYYY-MM-DD"
    close_price  REAL NOT NULL,
    PRIMARY KEY (index_name, date)
);
CREATE INDEX idx_benchmark_date ON benchmark_data(date);
```

Listing 3: benchmark_data — index close prices

Both yfinance-sourced indices and NSE-archive-sourced indices are stored in this single table; the `index_name` column holds the display name used throughout the UI.

Table: `fund_tags`

```
CREATE TABLE fund_tags (
    scheme_code INTEGER NOT NULL,
    tag         TEXT    NOT NULL,
    PRIMARY KEY (scheme_code, tag),
    FOREIGN KEY (scheme_code) REFERENCES funds(scheme_code)
```

```sql
);
CREATE INDEX idx_fund_tags_tag  ON fund_tags(tag);
CREATE INDEX idx_fund_tags_code ON fund_tags(scheme_code);
```

Listing 4: fund_tags — sector/style taxonomy

Tags are assigned automatically by the tagger rule engine (see Section 15.6) and can be overridden per-fund via the Data Management page.

Table: `update_log`

```sql
CREATE TABLE update_log (
    id              INTEGER PRIMARY KEY AUTOINCREMENT,
    update_type     TEXT    NOT NULL,   -- "nav", "benchmark", etc.
    started_at      TEXT    NOT NULL,
    finished_at     TEXT,
    schemes_updated INTEGER DEFAULT 0,
    status          TEXT    DEFAULT 'running'  -- 'completed' | 'error'
);
```

Listing 5: update_log — audit trail

Every update run writes a row on start and updates it on completion. The Data Management page reads the latest row to display last-update status and timestamp.

Data Pipeline

Data enters the system in four stages, each handled by a function in `fetcher.py`.

AMFI NAVAll.txt	⟶	mfapi.in REST API	⟶	SQLite `nav_history`
NSE Archive CSV	⟶	NSE bulk ZIP downloads	⟶	SQLite `benchmark_data`
Yahoo Finance	⟶	yfinance Python library	⟶	SQLite `benchmark_data`

Stage 1 — Scheme Discovery (`fetch_scheme_list`)

AMFI publishes a plain-text file at `amfiindia.com/spages/NAVAll.txt` that lists every registered mutual fund scheme in India. Each line is semicolon-delimited:

`SchemeCode ; ISIN_Div ; ISIN_Growth ; SchemeName ; NAV ; Date`

Section headers of the form `Open Ended Schemes(Equity Scheme - Large Cap Fund)` identify the SEBI category for the schemes that follow. The parser (`parse_amfi_nav_file`) maintains a running `current_category` variable updated on each header line, then attaches it to every data record.

The filter `filter_equity_schemes` retains only Direct Growth plans in equity or ELSS categories. It rejects IDCW, dividend, payout, regular, institutional, and non-equity scheme names via a keyword exclusion list (`_DIRECT_GROWTH_EXCLUDE`).

Stage 2 — NAV History (`download_nav_for_schemes`)

Full NAV history for each scheme is fetched from the open `mfapi.in` REST API:

`GET https://api.mfapi.in/mf/{scheme_code}`

The API returns up to the full history in a single JSON response. Each record contains a date string in `dd-mm-yyyy` format which is converted to ISO-8601 (`yyyy-mm-dd`) before storage. The download is incremental: `get_last_nav_date` retrieves the most recent date already stored and only records newer than that date are inserted.

Rate limiting is enforced at `MFAPI_DELAY_SECONDS` (0.5 s) between requests and database commits happen every `MFAPI_BATCH_SIZE` (50) schemes to balance throughput against memory usage. A three-attempt exponential back-off (delays: 2 s, 4 s, 8 s) handles transient 5xx errors and timeouts.

Stage 3a — Benchmark Prices via yfinance (`fetch_benchmark_data`)

The 23 indices in `BENCHMARKS` (Table 4) are downloaded via the `yfinance` library. For each index the latest stored date is retrieved and only the incremental tail is fetched using yfinance's `start` parameter. Close prices are stored under the index's display name (e.g. `"Nifty 50"`).

Stage 3b — NSE Archive Indices (`fetch_nse_index_data`)

Seven indices — Nifty Smallcap 50/100/250, Nifty Microcap 250, Nifty Midcap 100, Nifty 500, and Nifty MidSmallcap 400 — are not available with meaningful history on yfinance. NSE publishes a bulk archive of all index closes in a daily CSV:

`archives.nseindia.com/content/indices/ind_close_all_DDMMYYYY.csv`

Each file is approximately 16 KB and contains every NSE index for that trading day. The downloader builds a reverse lookup from lowercase CSV column names to display names, finds the earliest date not yet stored, generates the list of weekday dates through today, downloads each file with a 50 ms inter-request delay, and performs bulk inserts. Files for dates before an index existed in the archive simply produce zero records for that index with no error.

Index	Source	History from
Nifty 50, Nifty Bank, Nifty IT, . . .	yfinance	Varies (2000+)
Nifty Smallcap 50/100/250	NSE archive CSV	Nov 2015
Nifty Microcap 250	NSE archive CSV	Jan 2016
Nifty Midcap 100	NSE archive CSV	Apr 2016
Nifty 500	NSE archive CSV	Jan 2013
Nifty MidSmallcap 400	NSE archive CSV	Nov 2015

Incremental Update Logic

All fetchers follow the same pattern: read the watermark (`MAX(date)`) for each series from the DB, then request only newer data. First-time population downloads the full history; subsequent runs fetch only the delta. The `update.py` script orchestrates Stages 1–3 in sequence and can be run on-demand or scheduled via cron.

Application Layer — Streamlit Pages

The UI is a single-file Streamlit application. Navigation is via a sidebar radio widget with five pages.

Dashboard

Purpose: Cross-fund returns heatmap with filtering.

Sidebar controls:

- Sector/style multiselect (from `fund_tags`)
- SEBI category multiselect
- Fund filter: Show all / Exclude funds / Pick funds
- Fund house, min history, and nav-count filters

Main area: A styled `pd.DataFrame` where each cell shows a trailing return period (6M, 1Y, 2Y, 3Y, 5Y) for each fund. Color coding uses `_signed_gradient_col`: the RdYlGn colormap is applied symmetrically around zero so any positive return is green and any negative return is red, per column.

Fund Analysis

Purpose: Deep dive into a single fund with a full analytics suite.

Sidebar controls: Fund picker, benchmark picker (with `TAG_BENCHMARK` auto-suggestion), date-range slider.

Tabs:

1. **NAV Chart** — line chart with benchmark overlay
2. **Returns** — trailing return table, calendar year bar chart, SIP XIRR summary
3. **Risk Metrics** — volatility, Sharpe, Sortino, Calmar, Martin, Ulcer, VaR, CVaR, downside deviation, max drawdown
4. **Drawdown** — drawdown time series, drawdown statistics
5. **Benchmark-Relative** — alpha, beta, R^2, tracking error, information ratio, capture ratios, batting average; rolling alpha and beta charts; rolling Sharpe/Sortino comparison

All 45 metrics in the Risk tab are computed by a module-level `@st.cache_data`-decorated wrapper around `compute_fund_analytics` so that changing a rolling-window widget does not re-run the full XIRR root-finding computation.

Fund Comparison

Purpose: Side-by-side comparison of up to 10 user-selected funds.

Tabs:

1. **NAV Chart** — rebased to 100 from common start date
2. **Returns** — returns heatmap across funds and periods
3. **Rolling** — rolling return and rolling Sharpe over user-selected window
4. **Full Table** — all trailing return columns sortable
5. **Risk** — volatility, Sharpe, max drawdown, peer quartile rank for each fund

Sector & Style

Purpose: Peer group comparison within a tag (sector or style).

Sidebar controls:

- Tag picker (all tags from `fund_tags`)
- Benchmark picker with `TAG_BENCHMARK` auto-selection
- Fund filter: Show all / Exclude funds / Pick funds

Tabs: Returns heatmap, risk metrics comparison, peer distribution charts, tag management (assign/remove tags per fund).

Data Management

Purpose: Data pipeline controls and database statistics.

Steps exposed in UI:

1. Refresh scheme list from AMFI
2. Download NAV history (full or incremental) with live progress
3. Download yfinance benchmark data
4. Download NSE archive indices (Smallcap, Microcap, Midcap 100, etc.)
5. Re-run auto-tagging
6. Daily update (all steps in sequence)

DB statistics (total funds, NAV records, benchmarks loaded, last update time) are displayed on the page.

Fund Tagging Rule Engine

Every fund in the database receives one or more tags via `auto_tag_all_funds()`. The engine operates in two passes:

Pass 1 — Style from SEBI category. Non-sectoral equity funds are uniquely identified by their SEBI category string. The `CATEGORY_TAG` dict maps each category directly to a style tag:

SEBI Category	Tag
Equity Scheme - Large Cap Fund	Large Cap
Equity Scheme - Mid Cap Fund	Mid Cap
Equity Scheme - Small Cap Fund	Small Cap
Equity Scheme - Large & Mid Cap Fund	Large & Mid Cap
Equity Scheme - Multi Cap Fund	Multi Cap
Equity Scheme - Flexi Cap Fund	Flexi Cap
Equity Scheme - Focused Fund	Focused
Equity Scheme - Value Fund	Value
Equity Scheme - Contra Fund	Contra
Equity Scheme - ELSS	ELSS

If a match is found, the function returns immediately — no name-based matching is needed.

Pass 2 — Theme from fund name (sectoral/thematic only). For funds in the `"Equity Scheme - Sectoral/ Thematic"` SEBI category, the engine compares the lowercased fund name against an ordered list of `(tag, keywords)` rules (`THEME_RULES`). The *first* matching rule wins, so rule ordering is critical. Rules for highly specific or easily conflated themes (e.g. Business Cycle and Momentum) are placed above the broader Quant & Factor rule to prevent misclassification by AMC name tokens like "Quant".

If no rule matches, the fund receives the tag `"Other"`. Funds can match at most one theme (the `break` after the first match enforces this).

Tag	Sample keywords
Banking & Financial Services	banking, financial services, bfsi
Technology	technology, digital india, information technology
Healthcare	pharma, healthcare, health and wellness
Infrastructure	infrastructure, build india, roads, railways
Consumption & FMCG	consumption, fmcg, consumer durable
Manufacturing	manufacturing, make in india
Energy & Resources	energy, natural resources, new energy
PSU	psu fund, public sector
Defence	defence, defense, strategic
ESG & Ethical	esg, ethical fund, best-in-class
Business Cycle	business cycle
Momentum	momentum fund, active momentum
Quant & Factor	quant fund, multi-factor, smart beta
International	global equity, japan equity, emerging market
...	(24 tags total)

Caching and Performance

Streamlit `@st.cache_data`. All expensive computations are wrapped in module-level `@st.cache_data`-decorated functions. The cache key is the function arguments tuple; Streamlit serialises `pd.Series` inputs via their hash. The critical module-level wrappers are:

- `_cached_fund_analytics(nav, bench)` — wraps `compute_fund_analytics`; avoids recomputing 45 metrics (including four XIRR root-finding operations) when the user changes a widget that does not affect the fund or date range.
- `_cached_rolling_sip(scheme_code, yrs)` — wraps `rolling_sip_xirr`; the rolling 3-year SIP XIRR is an $O(N)$ scan of XIRR calls and can take 2–5 s for funds with 10+ year histories.
- `load_nav_as_series(scheme_code)` — database read cached by scheme code; invalidated on `st.cache_data.clear()` after any data download.

Important: cached functions must be defined at module scope, not inside `with` blocks or nested function bodies. Locally-defined decorated functions receive a new identity on each Streamlit rerun, making the cache ineffective.

SQLite WAL mode. The database is opened with `PRAGMA journal_mode=WAL` and `PRAGMA synchronous=NORMAL` on every connection. WAL mode allows concurrent reads while a write is in progress, which prevents the Streamlit UI from blocking during background data downloads.

Incremental fetching. Neither NAV history nor benchmark prices are ever re-downloaded in full after the initial population. Every fetch call reads `MAX(date)` from the DB and requests only the tail. For daily updates this means each scheme requires a single API call returning typically 1–5 records.

Configuration Reference

All tuneable constants live in `config.py`. Table 3 documents every constant.

Table 3: config.py constants

Constant	Default	Purpose
`DB_PATH`	`./mf_data.db`	SQLite file path
`AMFI_NAV_URL`	amfiindia.com/.	AMFI scheme list endpoint
`MFAPI_BASE_URL`	api.mfapi.in/mf	NAV history base URL
`MFAPI_DELAY_SECONDS`	0.5	Inter-request delay for mfapi.in
`MFAPI_BATCH_SIZE`	50	DB commit frequency during NAV download
`RISK_FREE_RATE`	0.065	Annualised risk-free rate (India 91-day T-bill)
`TRADING_DAYS_PER_YEAR`	252	Used for annualisation of daily statistics
`BENCHMARKS`	23 entries	Dict of display name → yfinance ticker
`NSE_DIRECT_INDICES`	7 entries	Dict of display name → NSE CSV column name
`NSE_ARCHIVE_URL`	archives.nseindia.com/...	URL template for NSE bulk CSV
`TAG_BENCHMARK`	∼30 entries	Preferred benchmark per sector/style tag
`EQUITY_CATEGORY_KEYWORDS`	2 entries	Substrings to detect equity/ELSS funds
`ROLLING_WINDOWS`	5 entries	Name → trading-day count (6M=126, . . . , 5Y=1260)
`VAR_CONFIDENCE_LEVELS`	[0.95, 0.99]	Confidence levels for VaR / CVaR

Benchmark Coverage

Table 4: Benchmark indices available in the platform

Index	Source	Category
Nifty 50	yfinance	Broad market
Nifty 100	yfinance	Broad market
BSE Sensex	yfinance	Broad market
Nifty Next 50	yfinance	Broad market
Nifty Midcap 50	yfinance	Cap segment
Nifty Midcap 150	yfinance	Cap segment
Nifty Smallcap 50	NSE archive	Cap segment

(continued)

Index	Source	Category
Nifty Smallcap 100	NSE archive	Cap segment
Nifty Smallcap 250	NSE archive	Cap segment
Nifty Microcap 250	NSE archive	Cap segment
Nifty Midcap 100	NSE archive	Cap segment
Nifty MidSmallcap 400	NSE archive	Cap segment
Nifty 500	NSE archive	Broad market
Nifty Bank	yfinance	Sector
Nifty PSU Bank	yfinance	Sector
Nifty Financial Svc	yfinance	Sector
Nifty IT	yfinance	Sector
Nifty FMCG	yfinance	Sector
Nifty Consumption	yfinance	Sector
Nifty Pharma	yfinance	Sector
Nifty Infrastructure	yfinance	Sector
Nifty Energy	yfinance	Sector
Nifty Metal	yfinance	Sector
Nifty Media	yfinance	Sector
Nifty Commodities	yfinance	Sector
Nifty Services	yfinance	Sector
Nifty PSE	yfinance	Sector
Nifty Auto	yfinance	Sector
Nifty Realty	yfinance	Sector
Nifty MNC	yfinance	Sector

Deployment and Setup

Requirements

- Python 3.11+ (tested on 3.14)
- Packages: `streamlit`, `pandas`, `numpy`, `scipy`, `plotly`, `yfinance`, `requests`, `matplotlib`
- Optional: `schedule` (for `update.py -schedule` mode)

First-time setup

1. `pip install -r requirements.txt`
2. `python update.py` — populates scheme list and downloads NAV history for all equity Direct Growth funds (≈3–6 h for the initial full download of ∼1,700 schemes)
3. `streamlit run app.py` — launches the web UI on `localhost:8501`
4. In the Data Management page: download yfinance benchmarks, then NSE archive indices, then run auto-tagging

Daily update

The simplest approach is a `cron` job:

```
0 22 * * 1-5  cd /path/to/mf_analytics && python update.py >>
   update.log 2>&1
```

Listing 6: Cron entry for weekday updates at 22:00 IST

Alternatively, the built-in scheduler mode keeps a persistent process running:

```
python update.py --schedule --time 22:00
```

Listing 7: Persistent scheduler mode

The Streamlit UI also exposes a "Daily Update" button in the Data Management page for on-demand manual updates with a live progress indicator.

Data flow summary

Action	Duration (first run)	Duration (daily)
Scheme list refresh	<5 s	<5 s
NAV history download	3–6 h (∼1700 schemes)	5–15 min
yfinance benchmarks	<2 min	<30 s
NSE archive indices	15–30 min (3453 files)	<5 min
Auto-tagging	<1 s	<1 s

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